12 Below is a faulty argument that appears to show that the gradient of the curve $y = x^2$ at the point (3, 9) is 1.

> Consider the chord joining (3, 9) to the point $(3+h, (3+h)^2)$
>
> The gradient is $\frac{(3+h)^2 - 9}{h} = \frac{6h + h^2}{h}$
>
> When $h = 0$ the gradient is $\frac{0}{0}$ so the gradient of the curve is 1

(a) Identify a fault in the argument. **[1]**

(b) Write a valid first principles argument leading to the correct value for the gradient at (3, 9). **[3]**

(c) Find the equation of the normal to the curve at the point (3, 9). **[2]**

12(a)

12(b)

12(c)